

Internal Standard Method to Measure Nickel Nitrate Concentration with Fluorescence Spectroscopy

1 Learning Goals

- Learn the basics of internal standard method
- Determine the precision and accuracy of your analysis
- ***Verify your quantitative laboratory skills***

2 Introduction

In this experiment, you will apply the internal standard method to determine the unknown concentration of $\text{Ni}(\text{NO}_3)_2$ in solution by using absorbance spectroscopy. The internal standard method is important when the sample is contained in a matrix. In this experiment, $\text{Co}(\text{NO}_3)_2$ is chosen as the internal standard. The matrix consists of other chemicals that are present in the sample. In short, one fills (usually) five volumetric flasks each with the different concentration ratio of $\text{Ni}(\text{NO}_3)_2$ and $\text{Co}(\text{NO}_3)_2$. By plotting the signal ratio versus concentration ratio, a calibration curve is created. The measurement of the solution of unknown and internal standard provides the signal ratio, which can be used to obtain the concentration ratio. The concentration ratio can be used to calculate the unknown concentration.

3 Reagents

- ❖ Solution of $\text{Ni}(\text{NO}_3)_2$ (~1.000 M), you have your own concentration that is close to 1.000 M
- ❖ Solution of $\text{Co}(\text{NO}_3)_2$ (~1.000 M), you have your own concentration that is close to 1.000 M
- ❖ Solution with unknown ($\text{Ni}(\text{NO}_3)_2$) and internal standard ($\text{Co}(\text{NO}_3)_2$)
- ❖ **SOLVENT:** 0.01 M HNO_3

4 Procedures

4.1 Preparation of solutions

1. Prepare 100.0 mL ~1.000 M $\text{Ni}(\text{NO}_3)_2$. Clean and rinse one 100 mL volumetric flasks carefully with distilled water. Do the calculation and measure $\text{Ni}(\text{NO}_3)_2$ by analytical balance.
2. Prepare 100.0 mL ~1.000 M $\text{Co}(\text{NO}_3)_2$. Clean and rinse one 100 mL volumetric flasks carefully with distilled water. Do the calculation and measure $\text{Co}(\text{NO}_3)_2$ by analytical balance.
3. Prepare 5 internal standard solution with different ratio as shown in table below using five 50 mL volumetric flasks.

Volume of $\text{Ni}(\text{NO}_3)_2$, mL	5	10	15	20	25
Volume of $\text{Co}(\text{NO}_3)_2$, mL	25	20	15	10	5

4. Prepare $\sim 0.200 \text{ M}$ $\text{Ni}(\text{NO}_3)_2$ solution in a 50 mL volumetric flask by diluting the $\sim 1.000 \text{ M}$ $\text{Ni}(\text{NO}_3)_2$ solution.
5. Prepare $\sim 0.200 \text{ M}$ $\text{Co}(\text{NO}_3)_2$ solution in a 50 mL volumetric flask by diluting the $\sim 1.000 \text{ M}$ $\text{Co}(\text{NO}_3)_2$ solution.

4.2 Measurements by the Vernier Spectrophotometer

For this experiment we will use a Vernier Spectrometer, an instrument that can quickly measure the entire absorption spectrum of a sample. It is necessary to set the instrument's scale to where no light is absorbed and to where all the light is absorbed. This process is called "calibration".

Safe Handling and proper use of the Vernier LabQuest 2

*The Vernier LabQuest module and the SpectroVis Plus spectrophotometer are expensive pieces of equipment. Please take the following precautions while you use them. Treat the device as if it was your cellphone and **you had to pay the bill if you damaged it.***

- A. Treat the devices carefully. Place them on the bench top well away from the edge and do not walk around with them (except for getting a unit and storing them).*
- B. Keep liquids well away from the module. Organize your bench space such that your solutions are separated from your Vernier module. Do not use the module with non-aqueous solutions. Be extra careful when using acids and bases.*
- C. Use the stylus to touch the screen, rather than your fingers. You may transfer chemicals when using your fingers.*
- D. When done, switch off the device. Wipe off any liquids that were accidentally spilled on the module. Put the stylus back in the device. Put everything away in its proper place: LabQuest modules with LabQuest modules in one box, spectrometers with spectrometers in another box, etc.*
- E. Report any irregularities to your lab instructor.*

1. Obtain a LabQuest module and SpectroVis Plus spectrophotometer from your lab instructor. You also need a power supply and USB cable. The power supply plugs into the LabQuest and the cable connects the spectrophotometer with the LabQuest unit. The LabQuest should also have a small stylus in a slot on the back of the instrument.
2. Obtain ONE plastic cuvette – only touch the cuvettes on the very top $\frac{1}{4}$ or on the frosted sides. You do not want to get fingerprints on the clear windows where the light must travel.

3. Plug the USB cable from the spectrophotometer. Turn on the LabQuest unit (red on button on top left). The instrument will automatically take you to the Absorption App (red box with “USB: Abs”) if you have your spectrometer plug in before you turn on the console. If you don’t see the app started, you will need to start **LabQuest App** to get to the correct screen.
4. When you are in the app -- touch “**Sensors**” then “**Data Collection**” on the top of the screen. Enter the following settings:

Sample time: 200 ms
Wavelength smoothing: 3
Samples to average: 6
Wavelength range 380 to 700 nm
Then touch “ok”

5. Then go to **Sensors** then **Calibrate** then **USB: Spectrometer**.
6. Touch **ok** or just wait. You will get a new screen that tells you the spectrometer is warming up. Wait for the spectrometer to complete the warm up of the lamp.
7. After the instrument warms up, fill one cuvette with **0.01 M HNO₃**, wipe it clean with a Kimwipe, and place it in the sample chamber. Make sure the cuvette is positioned such that the clear windows are aligned with the white arrow and white lamp on the instrument (that is the direction the light will travel). Touch **finish calibration** on the screen.
8. The instrument will take ~ 5 seconds to calibrate itself (it is figuring how the signal from 100% of the light traveling through). When it says “calibration completed”. Touch **ok**.
9. The screen will go back to the original screen in the app (with a multi-colored background). It may also default immediately into a measurement (will see red box at bottom left rather than green arrow). It may also go back to the screen with the red box (“USB: Abs”). If the instrument didn’t start, start a measurement with only the water in the cuvette by touching the **green arrow**. Stop the measurement by touching the **red box in the lower left**. A very noisy flat line centered around 0 should appear on the graph. The magnitude of the noise should not be larger than 0.0070. If it is, you need to recalibrate (see step 10).
10. If your line in step 9 was flat and centered around 0, you are ready to measure.

Measure the absorbance of the background at least ten times at the maximum absorbance wavelength that you previously measured.

1. After each measurement discard the solution in the cuvette and replace it with fresh solution.
2. In this experiment, the background is 0.01 M **HNO₃**.
3. These measurements should be very similar, within 1-2% of each other. If your variability is larger than this, check the calibration settings on the UV-Vis

- Record each absorbance measurement in a table in your lab notebook.

Determine the wavelength of maximum absorbance for $\text{Ni}(\text{NO}_3)_2$

- Fill the cuvette 2/3 full with $\sim 0.200 \text{ M Ni}(\text{NO}_3)_2$ solution. Place the cuvette into the spectrophotometer such that the clear sides are **aligned with the white arrow**. Touch the **green arrow**. You may discard the previous run by touching **discard**. Within a second, the absorption spectrum of the solution should appear on the screen. Touch the **red box** to stop the measurement. Note: if you discard the previous data, your new spectrum will be labeled run 1. If you choose to store the previous data, your new spectrum will be labeled run 2. A subsequent run will be named run 3, etc. If you store, you can at any time revisit your runs by touching the run # box.
- Move the line cursor on the graph to the **peak** of the absorption curve. Record the wavelength (the x value, shown on the right side of the screen, in nm). The x value will be in nm (don't mistakenly write down the y-value). **You will now do all your measurements for $\text{Ni}(\text{NO}_3)_2$ at that absorbance wavelength.**

Determine the wavelength of maximum absorbance for $\text{Co}(\text{NO}_3)_2$

- Rinse the cuvette with DI water 3 times and then $\sim 0.200 \text{ M Co}(\text{NO}_3)_2$ solution three times. Fill the cuvette 2/3 full with $\sim 0.200 \text{ M Co}(\text{NO}_3)_2$ solution. Place the cuvette into the spectrophotometer such that the clear sides are **aligned with the white arrow**. Touch the **green arrow**. Within a second, the absorption spectrum of the solution should appear on the screen. Touch the **red box** to stop the measurement. **Note: Store your data from this point, don't discard any data. You can at any time revisit your runs by touching the run # box.**
- Move the line cursor on the graph to the **peak** of the absorption curve. Record the wavelength (the x value, shown on the right side of the screen, in nm). The x value will be in nm (don't mistakenly write down the y-value). **You will now do all your measurements for $\text{Co}(\text{NO}_3)_2$ at that absorbance wavelength.**

Measurement of the absorbance of each internal standard solution

- Measure the maximum absorbance of each of the standard addition solutions that you prepared **three times**.
 - After each measurement discard the solution in the cuvette and replace it with fresh solution.
 - Record each absorbance measurement in a table in your lab notebook.
- Measure the absorbance of the unknown solution with the internal standard that was provided to you at least seven times.

- After each measurement discard the solution in the cuvette and replace it with fresh solution.
 - Record each absorbance measurement in a table in your lab notebook.
3. After you have completed all of your measurements, discard all of your waste in the sink. Clean your

5 Cleanup

- Dispose of all other wastes in the designated waste container.
- Clean and rinse all the glassware that you have used and place it on the drying rack. Place volumetric pipettes in the oven (**careful: oven is hot!**).
- Wipe your bench space clean.

6 Data Analysis and Presentation of Results in Lab Reports

Analysis of blank measurements

- Measure the average signal and standard deviation of the ten blank measurements.
- *You will need to subtract the signal of the blank from all of your measurements once you've determined the average.*

Plot of calibration curve

- Plot the absorbance as a function of signal ratio versus concentration ratio.
- Perform a linear regression of the x,y scatter data and show the trend line on the graph.
- Using the LINEST function in Excel, determine the slope, intercept, and uncertainty in the slope and intercept. Report these values clearly in your lab write-up.
- Label this as Figure 1 in your lab report, and provide a description of the figure. The figure caption should provide enough detail to understand the result of the experiment.

Unknown calculation

Obtain the true value of the $\text{Ni}(\text{NO}_3)_2$ concentration of the unknown from your instructor and calculate the absolute error and relative error

- Provide all the information in a table. Pay attention to units and significant figures.

Concentration of $\text{Ni}(\text{NO}_3)_2$ (M)	Standard deviation of concentration (M)	RSD of concentration (%)	Absolute error (M)	Relative error (%)
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6.1 Questions

1. Based on your graph, what can you say about the quality of your internal standard curve? Explain your answer.
2. Was your measurement accurate? Explain your answer. Assume that a relative error of equal to or less than 5% is good, equal to or less than 10% is acceptable, and larger than 10% is not acceptable. Explain what may have caused the accuracy to be unacceptable, if it was, and how it can be improved.
3. Was your measurement precise? Explain your answer. Assume that a relative standard deviation smaller than or equal to 3% is acceptable. Explain what may have caused the precision to be unacceptable, if it was, and how it can be improved.